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1. Topological Spaces and Continuous Maps

1.1. Mathematical Spaces

Recall

Given a real inner product $\langle \cdot, \cdot \rangle$ on a real vector space V , we can define a norm on V by $\|x\| = \sqrt{\langle x, x \rangle}$.

Given a norm on a real vector space V , we can define a metric on V by $d(x, y) = \|x - y\|$.

Given a metric d on a set X , we can define open sets in X as follows:

Definition 1.1.1

Given $a \in X$ and $r > 0$, $B(a, r) = B_X(a, r) = \{x \in X \mid d(a, x) < r\}$, then for $A \subseteq X$, we say that A is open (in X) when $\forall a \in A, \exists r > 0$ such that $B(a, r) \subseteq A$.

Note

Equivalently, A is open (in X) when $\forall a \in A, \exists b \in X, \exists r > 0$ such that $a \in B(b, r) \subseteq A$.

Remark

The set of open sets in the metric space X is called the metric topology on X . Open sets in a metric space satisfy the following:

1. $\emptyset$ and X are open.
2. The union of any set of open sets in X is open (in X).
3. The intersection of any finite set of open sets in X is open (in X).

Notation

1. For a set S of sets, the union of S is $\cup S := \{x \mid \exists A \in S, x \in A\}$.
2. For $S \neq \emptyset$, the intersection of S is $\cap S := \{x \mid \forall A \in S, x \in A\}$.
3. In the case that $S = \{A_k \mid k \in K\}$ where K is a set (called the index set), then
$$\cup S = \bigcup_{k \in K} A_k \quad \cap S = \bigcap_{k \in K} A_k.$$

Definition 1.1.2

Let X be a set. A topology on X is a set $\mathcal{T}$ of subsets of X such that

1. $\emptyset \in \mathcal{T}$ and $X \in \mathcal{T}$
2. $\mathcal{T}$ is closed under arbitrary unions
3. $\mathcal{T}$ is closed under finite intersections

Remark

By induction, to verify that (3) holds, it suffices to show that
 $A, B \in \mathcal{T} \Rightarrow A \cap B \in \mathcal{T}$.

Definition 1.1.3

When $\mathcal{T}$ is a topology on X , the elements of $\mathcal{T}$ are called the open sets in X . The complements (in X) are called the closed sets in X .

Example

X set. $\mathcal{T} = \{\emptyset, X\}$ is a topology. Moreover, it's called the trivial topology.
 Also, $P = P(X) = \{A | A \subseteq X\}$ is a topology on X . It's called the discrete topology.

Recall

When $f : X \rightarrow Y$ is a map between metric spaces, f is continuous $\Leftrightarrow f$ has the property that $f^{-1}(V)$ is open in X for every open set V in Y .

Definition 1.1.4

For a map $f : X \rightarrow Y$ between topological spaces, we say that f is continuous when $f^{-1}(V)$ is open in X for every open V in Y .

Example

Recall that if $f : A \subseteq \mathbb{R}^n \rightarrow B \subseteq \mathbb{R}^m$ is an elementary function, then f is continuous.

Definition 1.1.5

A topological space is a set X together with a topology $\mathcal{T}$.

Definition 1.1.6

Let $\mathcal{S}, \mathcal{T}$ be topologies on X with $\mathcal{S} \subseteq \mathcal{T}$, we say that $\mathcal{S}$ is coarser than $\mathcal{T}$ and that $\mathcal{T}$ is finer than $\mathcal{S}$. When $\mathcal{S} \subsetneq \mathcal{T}$, we say that $\mathcal{S}$ is strictly coarser than $\mathcal{T}$ and that $\mathcal{T}$ is strictly finer than $\mathcal{S}$.

Example

Consider $X = \emptyset$. $T \subseteq P(X) = \{\emptyset\} \Rightarrow T = \emptyset$ or $T = \{\emptyset\}$. But only $T = \{\emptyset\}$ is a topology since $X = \emptyset \notin T$.

Example

Consider $X = \{a\}$. $\{\emptyset, \{a\}\} = P(X) \supseteq T \in \{\emptyset, \{\emptyset\}, \{\{a\}\}, \{\emptyset, \{a\}\}\}$. But $\emptyset \notin \emptyset$, $\{a\} \notin \{\emptyset\}$ and $\emptyset \notin \{\{a\}\}$. So $T = \{\emptyset, \{a\}\}$ is the only remaining candidate. We check that it is closed under arbitrary unions and finite intersections to conclude that it is a topology on X .

Exercise 1.1.7

Find all topologies on $X = \{a, b\}$ and $X = \{a, b, c\}$.

Solution:

Let $X = \{a, b\}$. Then $T \subseteq P(\{a, b\})$ such that $\emptyset \in T$ and $\{a, b\} \in T$. So

$T \in P = \{\{\emptyset, \{a, b\}\}, \{\emptyset, \{a, b\}, \{a\}\}, \{\emptyset, \{a, b\}, \{b\}\}, \{\emptyset, \{a, b\}, \{a\}, \{b\}\}\}$.

Note that each $T \in P$ is closed under arbitrary unions and finite intersections, so each one is a topology on X .

Similarly, $S_T = \{\{\emptyset, \{a, b, c\}\}, \{\emptyset, \{a, b, c\}, \{a\}\}, \{\emptyset, \{a, b, c\}, \{b\}\}, \{\emptyset, \{a, b, c\}, \{c\}\}, \dots\}$ for $X = \{a, b, c\}$.

Definition 1.1.8

Let X be a topological space and let $A \subseteq X$.

1. The interior of A (in X), denoted by A^0 , or by $\text{int}(A)$, or by $\text{int}_X(A)$, is the union (of the set) of all open sets in X , which are contained in A .
2. The closure of A (in X), denoted by $\overline{A}$, or by $Cl(A)$, or by $Cl_X(A)$, is the intersection of (the set of) all the closed sets in X , which contain A .
3. The boundary of A (in X), denoted by δA , or by $\delta_X A$, or by $Bd(A)$, or by $Bd_X(A)$, given by $\delta A = \overline{A} \setminus A^0 = \{x \mid x \in \overline{A}, x \notin A^0\}$.

Note

The set of closed sets in a topological space is closed under arbitrary intersection and under finite unions. So, in particular, $\emptyset$ and X are both closed.

Theorem 1.1.9

Let X be a topological space and let $A \subseteq X$. Then

1. A^0 is open in X , and it is the largest open set in X , which is contained in A . Also, $\overline{A}$ is closed in X , and $\overline{A}$ is the smallest closed set in X , which contains A .
2. A is open $\Leftrightarrow A = A^0$. Moreover, A is closed $\Leftrightarrow A = \overline{A}$
3. $(A^0)^0 = A^0$ and $\overline{\overline{A}} = \overline{A}$

Proof:

1. A^0 is a union of open sets in X , so it is open in X . Moreover, there cannot exist open B in X such that $A^0 \subsetneq B \subseteq A$, since A^0 is the union of all open sets in X contained

in A , so $B \subseteq A^0$.

Similarly, $X \setminus \overline{A}$ is a union of open sets in X , so it is open in X . Hence, $\overline{A}$ is open.

Moreover, there cannot exist closed B in X such that $A \subseteq B \subsetneq \overline{A}$ because $\overline{A}$ is the intersection of all closed sets in X containing A , so $\overline{A} \subseteq B$.

2. $A \text{ open} \Rightarrow A^0 = A$ since A^0 is the largest open set contained in A and $A \subseteq A^0$.
 $A^0 = A \Rightarrow A \text{ open}$ since A^0 is open from part (1).
 $A \text{ closed} \Rightarrow \overline{A} = A$ since $\overline{A}$ is the smallest closed set containing A and $A \subseteq \overline{A}$.
 $\overline{A} = A \Rightarrow A \text{ closed}$ since $\overline{A}$ is closed.
3. $(A^0)^0 = A^0$ since A^0 is open and since $(A^0)^0$ is the largest open set contained in A^0 .
 Moreover, $\overline{\overline{A}} = \overline{A}$ since $\overline{A}$ is closed and since $\overline{\overline{A}}$ is the smallest closed set containing $\overline{A}$.

□

Definition 1.1.10

Let X be a topological space and let $A \subseteq X$ with $a \in A$.

1. We say that a is an interior point of A when $a \in A$ and there exists an open set U in X with $a \in U \subseteq A$.
2. We say that a is a limit point of A when for every open set U in X with $a \in U$, we have $(U \setminus \{a\}) \cap A \neq \emptyset$.

The set of limit points of A in X is denoted by A^1 , or by $\text{Lim}(A)$.

3. We say that a is a boundary point of A (in X) when for every open set U in X with $a \in U$, we have $U \cap A \neq \emptyset$ and $U \cap A^c \neq \emptyset$ (where $A^c = X \setminus A$).

Theorem 1.1.11

Let X be a topological space and let $A \subseteq X$.

1. A^0 is equal to the set of all interior points of A in X .
2. For $a \in X$, $a \in A^1 \Leftrightarrow a \in \overline{A} \setminus \{a\}$.
3. A is closed $\Leftrightarrow A^1 \subseteq A$.
4. $\overline{A} = A \cup A^1$.
5. $\overline{A}$ is the disjoint union $\overline{A} = A^0 \cup \delta A$.
6. δA is equal to the set of boundary points of A in X .

Proof:

1. Let A_I denote the set of interior points in A . $a \in A^0 \Rightarrow a \in U \subseteq A$, where U is open. So $a \in A_I$. If $a \in A_I$, then there exists open U such that $a \in U \subseteq A$. So $a \in A^0$. Hence, $A^0 = A_I$.
2. Let $a \in X$. $a \in A^1$. Let C be a closed set such that $A \setminus \{a\} \subseteq C$.
- 3.

□

1.2. Topological bases

Theorem 1.2.1

Let X be a topological space. The intersection of any set of topologies on X is also a topology on X .

Proof:

Let R be a set of topologies on X . When $R = \emptyset$, since the elements of R lie in the power set $P(X)$, we take $\cap R$ to be the set $P(X)$, which is the discrete topology on X .

Suppose $R \neq \emptyset$. $\forall A \in R, \emptyset \in A$ since A is a topology, so $\emptyset \in \cap R$. Similarly,

$\forall A \in R, X \in A$ since A is a topology on X , so $X \in \cap R$.

Now, let $S \subseteq \cap R$. For $A \in R$, since $S \subseteq \cap R$ and $A \in R$, then $S \subseteq A$. But A is a topology so it is closed under arbitrary unions. Therefore, $\cup S \in A$. But $A \in R$ is arbitrary, so $\cup S \in \cap R$. Hence, $\cap R$ is closed under arbitrary unions.

Finally, let $T_1, T_2 \in \cap R$. For $A \in R$, since $T_1 \in \cap R$ and $A \in R$, then $T_1 \in A$. Similarly, $T_2 \in A$. But A is a topology, so $T_1 \cap T_2 \in A$. However, $A \in R$ was arbitrary, so $T_1 \cap T_2 \in \cap R$. Hence, $\cap R$ is closed under finite intersection. $\square$

Note

Given a set X , the union of any set of topologies is not a topology on X in general.

Corollary 1.2.2

When X is a set and $\mathcal{S}$ is any set of subsets of X (that is, $\mathcal{S} \subseteq P(X)$), there exists a unique smallest (coarsest) topology $\mathcal{T}$ on X which contains $\mathcal{S}$. In particular, $\mathcal{T}$ is the intersection of (the set of) all topologies on X containing $\mathcal{S}$.

Note

This topology $\mathcal{T}$ is called the topology on X generated by $\mathcal{S}$.

Definition 1.2.3

Let X be a set. A basis of sets on X is a set $\mathcal{B}$ of subsets of X (so $\mathcal{B} \subseteq P(X)$) such that:

1. $\mathcal{B}$ covers X (that is, $\cup \mathcal{B} = X$)
2. $\forall C, D \in \mathcal{B}, \forall a \in C \cap D, \exists B \in \mathcal{B}, a \in B \subseteq C \cap D$.

Note

When $\mathcal{B}$ is a basis of sets in X and $\mathcal{T}$ is the topology on X generated by $\mathcal{B}$, we say that $\mathcal{B}$ is a basis for $\mathcal{T}$.

The elements in $\mathcal{B}$ are called the basic open sets in X .

Theorem 1.2.4 Characterization of Open Sets in terms of Basic Open Sets

Let X be a topological space, $\mathcal{B}$ be a basis for the topology on X .

1. For $A \subseteq X$, A is open in X if and only if A has the property that
 $\forall a \in A, \exists B \in \mathcal{B}, a \in B \subseteq A$.
2. The open sets in X are the unions of (the set of) elements in $\mathcal{B}$.

Proof:

Let $\mathcal{T}$ be the topology on X generated by the basis $\mathcal{B}$. Let $\mathcal{S}$ be the set of all sets $A \subseteq X$ with the property $\forall a \in A, \exists B \in \mathcal{B}, a \in B \subseteq A$ ($*$) and let $\mathcal{R}$ be the set of (arbitrary) unions of (sets of) elements in $\mathcal{B}$.

Recall that $\mathcal{T}$ is the intersection of the set of all topologies on X which contain $\mathcal{B}$. Note that $\mathcal{S}$ contains $\mathcal{B}$, since

$$\begin{aligned} A \in \mathcal{B} &\Rightarrow \forall C, D \subseteq X, \forall a \in C \cap D, \exists B \in \mathcal{B}, a \in B \subseteq C \cap D \\ &\Rightarrow \forall A \subseteq X, \forall a \in A, \exists B \in \mathcal{B}, a \in B \subseteq A \\ &\Rightarrow A \in \mathcal{S} \end{aligned}$$

Now, let us show that $\mathcal{S}$ is a topology on X .

We have $\emptyset \in \mathcal{S}$ vacuously and $X \in \mathcal{S}$ because $\mathcal{B}$ covers X (given $a \in X$, we can choose $B \in \mathcal{B}$ with $a \in B$). When $U_k \in \mathcal{S}$ for every $k \in K$ (where K is any index set), we need to show that $\bigcup_{k \in K} U_k \in \mathcal{S}$.

Let $a \in \bigcup_{k \in K} U_k$. Choose $l \in K$ so that $a \in U_l$. Since $U_l \in \mathcal{S}$ (so U_l satisfies $*$), we can choose $B \in \mathcal{B}$ so that $a \in B \subseteq U_l$. Since $U_l \subseteq \bigcup_{k \in K} U_k$, we have $a \in B \subseteq \bigcup_{k \in K} U_k$. Thus $\bigcup_{k \in K} U_k$ satisfies $*$, hence $\bigcup_{k \in K} U_k \in \mathcal{S}$, as required.

Suppose $U, V \in \mathcal{S}$ (so U and V satisfy $*$).

Let $a \in U \cap V$. Since $U \in \mathcal{S}$, we can choose $C \in \mathcal{B}$ with $a \in C \subseteq U$.

Since $V \in \mathcal{S}$, then we can choose $D \in \mathcal{B}$ with $a \in D \subseteq V$. Since $\mathcal{B}$ is a basis, $C, D \in \mathcal{B}$ and $a \in C \cap D$, we can choose $B \in \mathcal{B}$ with $a \in B \subseteq C \cap D$.

Then we have $a \in B \subseteq C \cap D \subseteq U \cap V$. Thus $U \cap V$ satisfies $*$, so that $U \cap V \in \mathcal{S}$, as required. Thus $\mathcal{S}$ is a topology on X containing $\mathcal{B}$, hence $\mathcal{T} \subseteq \mathcal{S}$.

Let us show that $\mathcal{S} \subseteq \mathcal{R}$. Let $U \in \mathcal{S}$ (so U satisfies $*$). For each $a \in U$, choose $B_a \in \mathcal{B}$ with $a \in B_a \subseteq U$. Then we have $U = \bigcup_{a \in U} B_a \in \mathcal{R}$. Thus $\mathcal{S} \subseteq \mathcal{R}$.

Finally, note that $\mathcal{R} \subseteq \mathcal{T}$ because if $U = \bigcup_{k \in K} B_k$ with $B_k \in \mathcal{B}$, then each $B_k \in \mathcal{T}$ and $\mathcal{T}$ is a topology, so $U = \bigcup_{k \in K} B_k \in \mathcal{T}$.

Therefore, we have that $\mathcal{T} \subseteq \mathcal{S} \subseteq \mathcal{R} \subseteq \mathcal{T}$, so $\mathcal{R} = \mathcal{S} = \mathcal{T}$. But $\mathcal{S} = \mathcal{T}$ is equivalent to (1) and $\mathcal{R} = \mathcal{T}$ is equivalent to (2). □

Theorem 1.2.5 Characterization of a Basis in terms of Open Sets

Let X be a topological space with topology $\mathcal{T}$ and let $\mathcal{B} \subseteq \mathcal{T}$. Then $\mathcal{B}$ is a basis for $\mathcal{T}$ iff $\forall U \in \mathcal{T}, \forall a \in U, \exists B \in \mathcal{B}, a \in B \subseteq U$ ($*$).

Proof: If $\mathcal{B}$ is a basis for $\mathcal{T}$, then $(*)$ holds by part (1) of the previous theorem.

Suppose that $(*)$ holds. Let's show that $\mathcal{B}$ is a basis of sets in X .

Note that $\mathcal{B}$ covers X since, taking $U = X$ in $(*)$, we have

$\forall a \in X, \exists B \in \mathcal{B}, a \in B (\subseteq X)$. Also note that given $C, D \in \mathcal{B}$ and $a \in C \cap D$, then by taking $U = C \cap D$ in $(*)$ (noting that $C, D \in \mathcal{B} \subseteq \mathcal{T}$, so that $U = C \cap D \in \mathcal{T}$), we can choose $B \in \mathcal{B}$ with $a \in B \subseteq C \cap D$. Thus, $\mathcal{B}$ is a basis of sets in X .

It remains to show that $\mathcal{T}$ is the topology generated by $\mathcal{B}$.

Let $\mathcal{S}$ be the topology generated by $\mathcal{B}$. By part (2) of the previous theorem, $\mathcal{S}$ is the set of all unions of elements in $\mathcal{B}$. Also, $\mathcal{S}$ is the smallest topology which contains $\mathcal{B}$. Since $\mathcal{B} \subseteq \mathcal{T}$, and $\mathcal{T}$ is a topology, $\mathcal{S} \subseteq \mathcal{T}$.

Also, we have $\mathcal{T} \subseteq \mathcal{S}$ because given $U \in \mathcal{T}$, by property $(*)$, for each $a \in U$, we can choose $B_a \in \mathcal{B}$ with $a \in B_a \subseteq U$, and then we have $\bigcup_{a \in U} B_a \in \mathcal{S}$ since it is a union of elements in $\mathcal{B}$. □

Example

When X is a metric space, the set $\mathcal{B}$ of all open balls in X is a basis for the metric topology on X .

Remark

We can use a basis for testing for various topological properties:

1. When X is a topological space, $\mathcal{B}$ is a basis for the topology on X , and $S \subseteq X$ and $a \in X$, then $a \in A^0 \Leftrightarrow \exists B \in \mathcal{B}$ with $a \in B \subseteq A$
2. $a \in \overline{A} \Leftrightarrow \forall B \in \mathcal{B}$ with $a \in B, B \cap A \neq \emptyset$
3. $a \in A^1 \Leftrightarrow \forall B \in \mathcal{B}$ with $a \in B, (B \setminus \{a\}) \cap A \neq \emptyset$
4. $a \in \delta A \Leftrightarrow \forall B \in \mathcal{B}$ with $a \in B, B \cap A \neq \emptyset \wedge B \cap (X \setminus A) \neq \emptyset$

Definition 1.2.6

A topological space X is called Hausdorff when it has the property that for all $a, b \in X$, with $a \neq b$, there exist disjoint open sets U and V in X with $a \in U$ and $b \in V$.

Example

Metric spaces are Hausdorff.

Definition 1.2.7**Subspace Topology**

Let Y be a topological space with topology $\mathcal{S}$, and $X \subseteq Y$ be a subset. Let $\mathcal{T} = \{V \cap X \mid V \in \mathcal{S}\}$. Then $\mathcal{T}$ is a topology on X . Indeed, $\emptyset \in \mathcal{S}$, so $\emptyset \cap X = \emptyset \in \mathcal{T}$ and $Y \in \mathcal{S}$, so $Y \cap X = X \in \mathcal{T}$. If K is any index set and $U_k \in \mathcal{T}$ for each $k \in K$, then for each $k \in K$, we can choose $V_k \in \mathcal{S}$ such that $U_k = V_k \cap X$, and then we have

$$\bigcup_{k \in K} U_k = \bigcup_{k \in K} (V_k \cap X) = \left(\bigcup_{k \in K} V_k \right) \cap X \in \mathcal{T}$$

since

$$\bigcup_{k \in K} V_k \in \mathcal{S}$$

Similarly, when K is finite and $U_k \in \mathcal{T}$ for each $k \in K$, we have $\bigcap_{k \in K} U_k \in \mathcal{T}$. The topology $\mathcal{T}$ on X is called the subspace topology on X (inherited from the topology on Y).

Theorem 1.2.8

Let Y be topological space, $\mathcal{C}$ be a basis for the topology on Y and let $X \subseteq Y$ be a subset. Then the set

$$\mathcal{B} = \{C \cap X \mid C \in \mathcal{C}\}$$

is a basis for the subspace topology on X .

Proof:

First, we prove that $\mathcal{B}$ is a basis of sets on X . Observe that

$$\bigcup \mathcal{B} = \bigcup_{k \in K} \mathcal{B}_k = \bigcup_{k \in K} (C_k \cap X) = \left(\bigcup_{k \in K} C_k \right) \cap X = Y \cap X = X$$

since $\mathcal{C}$ is a basis on Y . Moreover, for $C, D \in \mathcal{B}$, let $a \in C \cap D$ be arbitrary. Then $\exists F \in \mathcal{C}, C = F \cap X$ and $\exists G \in \mathcal{C}, D = G \cap X$. So $a \in (F \cap G) \cap X \subseteq F \cap G$. Since $\mathcal{C}$ is a basis of sets on Y (since it is a basis for the topology on Y), then $\exists C_0 \in \mathcal{C}$ such that $a \in C_0 \subseteq F \cap G$. But $a \in X$, so $a \in C_0 \cap X \subseteq (F \cap G) \cap X = C \cap D$. But $C_0 \cap X \in \mathcal{B}$, and since $a \in C \cap D$ was arbitrary, then $\mathcal{B}$ satisfies property 2 of [Definition 1.2.3](#). So $\mathcal{B}$ is a basis of sets.

Now, we argue that $\mathcal{B}$ generates X . Let $\mathcal{S}$ be the subspace topology on X , $\mathcal{R}$ be the set of all topologies on X containing $\mathcal{B}$. We must show that $\bigcap \mathcal{R} = \mathcal{S}$.

$$\bigcap \mathcal{R} = \{A \mid \forall B \in \mathcal{R}, A \in B\}. \mathcal{S} = \{C \cap X \mid C \in \mathcal{C}\}.$$

□

Theorem 1.2.9

Let Z be a topological space. Let $Y \subseteq Z$ be a subspace (meaning that Y uses the subspace topology) and $X \subseteq Y$ be a subset. Then the subspace topology on X inherited from Y is equal to the subspace topology on X inherited from Z .

Proof: Exercise. □

Theorem 1.2.10

Let Y be a metric space (using the metric space topology) and let $X \subseteq Y$. Then the subspace topology on X (inherited from the topology on Y) is equal to the metric topology on X , using the metric on X obtained by restricting the metric on Y .

Proof: Exercise. □

1.3. Continuous Maps

Definition 1.3.1**Continuity at a Point**

Let X, Y be topological spaces. For $f : X \rightarrow Y$ and $a \in X$, we say that f is continuous at a when for every open sets $V \in Y$ with $f(a) \in V$, there exists an open set $U \in X$ with $a \in U \subseteq f^{-1}(V)$. We say that f is continuous (in or on X) when for every open set $V \in Y$, $f^{-1}(V)$ is open in X .

Definition 1.3.2**Homeomorphism**

A homeomorphism from X to Y is a bijective map $f : X \rightarrow Y$ such that both f and its inverse $f^{-1} : Y \rightarrow X$ are continuous.

Note

We say that X and Y are *homeomorphic*, and we write $X \cong Y$ when there exists a homeomorphism $f : X \rightarrow Y$ (and we remark that then $f^{-1} : Y \rightarrow X$ is also a homeomorphism).

Theorem 1.3.3

Constant maps and inclusion maps are continuous.

Proof: For $f : X \rightarrow Y$ given by $f(x) = c \in Y$ for all $x \in X$, when V is open in Y , $f^{-1}(V) = \begin{cases} X & \text{if } c \in V \\ \emptyset & \text{if } c \notin V \end{cases}$. When $X \subseteq Y$ is a subspace and $f : X \rightarrow Y$ is given by $f(x) = x$ for all $x \in X$, when V is open in Y ,

$$f^{-1}(V) = \{x \in X \mid f(x) \in V\} = \{x \in X \mid x \in V\} = V \cap X$$

which is open in X (when X is using the subspace topology). □

Remark

When Y is a topological space and $X \subseteq Y$ is a subset, we shall assume, unless indicated otherwise, that X used the subspace topology.

Theorem 1.3.4**Equivalent Definitions of Continuity**

Let $f : X \rightarrow Y$ be a map between topological spaces.

1. f is continuous (on X) $\Leftrightarrow f$ is continuous at every $a \in X$.
2. f is continuous (on X) $\Leftrightarrow$ for every closed set K in Y , $f^{-1}(K)$ is closed in X .
3. if $\mathcal{C}$ is a basis for the topology on Y , then f is continuous $\Leftrightarrow$ for every $C \in \mathcal{C}$ $f^{-1}(C)$ is open in X .

Proof:

1. $\Rightarrow$ Suppose f is continuous on X . Let $a \in X$. Let V be an open set in Y with $f(a) \in V$. Let $U = f^{-1}(V)$. Then $f^{-1}(V)$ is open since f is continuous and $a \in U \subseteq f^{-1}(V)$.

$\Leftarrow$ Suppose, conversely, that f is continuous at every $a \in X$. Let V be an open set in Y . For each $a \in f^{-1}(V)$, since f is continuous at a with $f(a) \in V$, we can choose an open set $U_a \subseteq X$ with $a \in U_a \subseteq f^{-1}(V)$. Then $f^{-1}(V) = \bigcup_{a \in f^{-1}(V)} U_a$, which is open in X since it is a union of open sets in X .

The rest is an Exercise □

Theorem 1.3.5**Composites of Continuous Functions are Continuous**

Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be continuous maps between topological spaces. Then the composite map $h = g \circ f : X \rightarrow Z$ is continuous.

Proof: Show that $h^{-1}(w) = f^{-1}(g^{-1}(w))$. □

Remark

Homeomorphism of topological spaces behaves like an equivalence relation on the class of topological spaces for topological spaces X, Y, Z .

$X \cong X$ (since the identity map $I : X \rightarrow X$ is a homeomorphism)

if $X \cong Y$, then $Y \cong X$ (when $f : X \rightarrow Y$ is a homeomorphism, so is $f^{-1} : Y \rightarrow X$).

if $X \cong Y$ and $Y \cong Z$, then $X \cong Z$ (indeed, if $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are homeomorphisms, then so is $h = g \circ f : X \rightarrow Z$ with $h^{-1} = (g \circ f)^{-1} = f^{-1} \circ g^{-1}$).

Example

$f : [0, 2\pi) \rightarrow S^1, f(t) = e^{it}$.

Theorem 1.3.6 **Restriction of Domain and Restriction or Extension of Codomain**

Let X, Y, Z be topological spaces. Suppose $f : X \rightarrow Y$ is continuous.

1. For any subspace $A \subseteq X$, then restriction $f : A \rightarrow Y$ is continuous.
2. If $Y \subseteq Z$ is a subspace, then $f : Y \rightarrow Z$ is continuous, and if $B \subseteq Y$ with $f(X) \subseteq B$, then $f : X \rightarrow B$ is continuous.

Proof: Exercise □

Lemma 1.3.7

The Glueing Lemma

Let $f : X \rightarrow Y$ be a map between topological spaces. Then

1. If $X = \bigcup_{k \in K} U_k$, where K is a non-empty set and each U_k is open in X , and if each restriction map $f|_{U_k} : U_k \rightarrow Y$ is continuous (where U_k is using the subspace topology), then f is continuous.
2. If $X = \bigcup_{k \in K} A_k$, where K is a non-empty finite set, and each A_k is closed in X , and if each restriction $f : A_k \rightarrow Y$ is continuous (where A_k is using the subspace topology), then f is continuous.

Proof: (1) Suppose $X = \bigcup_{k \in K} U_k$, where each U_k is open in X and suppose each restriction $g_k : U_k \rightarrow Y$ is continuous (where $g_k(x) = f(x)$ for all $x \in U_k$). Let $V \subseteq Y$ be open in Y . Note that

$$\begin{aligned} f^{-1}(V) &= \{x \in X \mid f(x) \in V\} \\ &= \bigcup_{k \in K} \{x \in U_k \mid f(x) \in V\} \\ &= \bigcup_{k \in K} \{x \in U_k \mid g_k(x) \in V\} \\ &= \bigcup_{k \in K} g_k^{-1}(V) \end{aligned}$$

For each $k \in K$, since $g_k : U_k \rightarrow Y$ is continuous, we know that $g_k^{-1}(V)$ is open in U_k . Since U_k is using the subspace topology, we can choose an open set W_k in X such that $g_k^{-1}(V) = W_k \cap U_k$. This is open in X since W_k and U_k are both open in X . Since $f^{-1}(V) = \bigcup_{k \in K} g_k^{-1}(V)$, it is a union of open sets in X , so it is open in X . Thus, f is continuous (on X).

(2) Exercise. First, show that for $f : X \rightarrow Y$, f is continuous (on X) $\Leftrightarrow f^{-1}(C)$ is closed in X for every closed C in Y . Then show that when $A \subseteq X \subseteq Y$, A is closed in X (using the subspace topology from Y) $\Leftrightarrow A = B \cap X$ for some closed set B in Y . □

Example

The map $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = \begin{cases} 2x & \text{if } x \leq 0 \\ x^2 & \text{if } x \geq 0 \end{cases}$ is continuous (by part (2) of the Glueing lemma)

Example

The circle $x^2 + y^2 = 1$ in $\mathbb{R}^2$ is homeomorphic to the ellipse

$$\frac{(x-a)^2}{A^2} + \frac{(y-b)^2}{B^2} = 1$$

in $\mathbb{R}^2$. Indeed, the map $\emptyset : S \rightarrow E$ given by $\emptyset(x, y) = \left(\frac{x-a}{A}, \frac{y-b}{B}\right)$ and its inverse $\varphi = \emptyset^{-1} : E \rightarrow S$ given by $\varphi(u, v) = (Au + a, Bv + b)$

Example

$\mathbb{R} \cong (-1, 1) \subseteq \mathbb{R}$. The map $g : \mathbb{R} \rightarrow (-1, 1)$ given by $g(x) = \arctan(x)$ is a homomorphism.

Example

The standard unit n -sphere in $\mathbb{R}^{n+1}$ is the set $\mathbb{S}^n = \{x \in \mathbb{R}^{n+1} \mid \|x\| = 1\}$, where p is the north pole $p = e_{n+1} = (0, \dots, 0, 1) \in \mathbb{S}^n$. We have $\mathbb{S}^n \setminus \{p\} \cong \mathbb{R}^n$. Indeed, we can use the stereographic projection map.

2. Examples of Topological Spaces

Definition 2.0.1

Let X be a set.

We sometimes write X_t to indicate that X is using the trivial topology $\mathcal{T}_t = \{\emptyset, X\}$.

We sometimes write X_d to indicate that X is using the discrete topology

$$\mathcal{T}_d = P(X) = \{A \mid A \subseteq X\}$$

We sometimes write X_c to indicate that X is using the co-finite topology

$$\mathcal{T}_c = \{A \subseteq X \mid A = \emptyset \vee X \setminus A \text{ is finite}\}$$

Remark

The closed sets in $\mathcal{T}_c$ are the finite sets and the set X . $\mathcal{T}_c$ is the coarsest topology where the singletons are closed.

Definition 2.0.2

When X is a metric space, we assume, unless otherwise indicated, that X uses the metric topology. Sometimes, we might write X_m to indicate that X is using the metric topology $\mathcal{T}_m$.

Definition 2.0.3

When Y is a topological space and $X \subseteq Y$ is a subset, we assume, unless otherwise indicated, that X uses the subspace topology. Sometimes, we might write X_s to indicate that X is using the subspace topology $\mathcal{T}_s$.

Note

When $X \subseteq \mathbb{R}^n$, we shall assume, unless otherwise indicated, that X is using $\mathcal{T}_m = \mathcal{T}_s$.

Definition 2.0.4

Let X be a set. A (strict, linear or total) **order** on X is a binary relation $<$ on X which satisfies:

1. For all $x, y \in X$, exactly one of the following holds:
 - a. $x < y$
 - b. $x = y$
 - c. $y < x$
2. For all $x, y, z \in X$, if $x < y$ and $y < z$, then $x < z$.

Definition 2.0.5

An **ordered set** is a set X with an order $<$. When X is an ordered set, we also define $\leq, >, \geq$ by stipulating that for all $x, y \in X$,

1. $x \leq y \iff x < y \vee x = y$
2. $x > y \iff y < x$
3. $x \geq y \iff y \leq x$

Remark

In an ordered set X , we can define an upper bound, a lower bound, the supremum, the infimum, the maximum and the minimum of a subset $A \subseteq X$.

Example

When X is an ordered set, and $A \subseteq X$ is a subset, $M = \max(A)$ when $M \in A$ with $M \geq x$ for all $x \in A$. $m = \min(A)$ when $m \in A$ with $m \leq x$ for all $x \in A$.

Definition 2.0.6

When X is an ordered set, we have the following subsets (which are called intervals in X): For $a, b \in X$ with $a < b$, we have

1. $(a, b) = \{x \in X \mid a < x < b\}$
2. $(a, b] = \{x \in X \mid a < x \leq b\}$
3. $[a, b) = \dots$
4. $[a, b] = \dots$

Example

In $\mathbb{Z}$ with its standard order, $(1, 5) = [2, 5) = (1, 4] = [2, 4] = \{2, 3, 4\}$.

Definition 2.0.7

Let X be an ordered set. The **order topology** on X is the topology $\mathcal{T}_o$, which is generated by the basis $\mathcal{B}_o$ of sets in X which consists of the following intervals:

1. (a, b) where $a, b \in X$ with $a < b$.
2. $(a, M]$ where $M = \max(X)$ and $a \in X$ with $a \neq M$ (in the case that X has a maximum M).
3. $[m, b)$ where $m = \min(X)$ and $b \in X$ with $b \neq m$ (in the case that X has a minimum).

Exercise 2.0.8

Verify that $\mathcal{B}_o$ is a basis of sets in X .

Proof: Clearly,

$$X = \bigcup_{a < b \in X} (a, b)$$

covers X if X has neither a max or a min. If X has a max and/or a min, include the intervals $(a, M]$ and $[m, b)$ in the union for any $a < M$ and/or $b > m$. □

Notation

We sometimes write X_o to indicate that X is using the order topology.

Example

$$\mathbb{R} = \mathbb{R}_o = \mathbb{R}_m.$$

Definition 2.0.9

Let X be an ordered set. The **lower limit** topology on X is the topology T_l generated by the basis of sets $\mathcal{B}_l$, which consists of intervals of the form $[a, b)$, or of the form $[c, \max(X)]$, where $a, b \in X$, $c \in X \setminus \{\max(X)\}$ with $a < b$.

Note

We sometimes write X_l to indicate that X is using the lower limit topology.

Example

On $\mathbb{R}$, $T_0 \subseteq T_m$ because sets in T_0 are open intervals and open intervals are open in $\mathbb{R}$'s metric topology. Also, $T_m \subseteq T_0$ by the following argument: open sets in $\mathbb{R}$'s metric topology are unions of open intervals because the set of open intervals is a basis of sets on $\mathbb{R}$ that generates T_m . But open intervals are in T_0 and T_0 is closed under arbitrary unions. So $T_m \subseteq T_0$ and we have that $T_0 = T_m$. However, T_l is not equal to T_m . Note that when $a, b \in \mathbb{R}$ with $a < b$,

$$(a, b) = \bigcup_{n=m}^{\infty} \left[a + \frac{1}{n}, b \right)$$

where $\frac{1}{m} < b - a$, which is open in $\mathcal{T}_l$, so we have $\mathcal{T}_0 \subseteq \mathcal{T}_l$.

Example

Let $X = (0, 1) \cup \{2\} \subseteq \mathbb{R}$. Note that $\mathcal{T}_0 \neq \mathcal{T}_m = \mathcal{T}_s$ on X , where X uses the standard order inherited from $\mathbb{R}$. For example, $\{2\}$ is open in X_m , since $B_X(2, \frac{1}{2}) = B_{\mathbb{R}}(2, \frac{1}{2}) \cap X = \{2\}$. But $\{2\}$ is not open in X_o because any open set in X_o which contains 2, must contain a basic open set $\mathcal{B}$ with $2 \in \mathcal{B} \subseteq U$, so it must contain a set of the form $\mathcal{B} = (1, 2]_X = (a, 1) \cup \{2\}$, where $a \in (0, 1)$. So they must include elements other than 2.

Example

When X is an ordered set, the dictionary (or lexicographic) order on X^2 is given by stipulating that

$$(a, b) < (c, d) \Leftrightarrow \begin{cases} a = c \text{ and } b < d \\ a < c \end{cases}$$

Note

On $\mathbb{R}^2$, the order topology $\mathcal{T}_0$ is not equal to the standard metric topology $\mathcal{T}_m$.

2.1. Products of Topological Spaces

Definition 2.1.1

Let K be a non-empty set and let X_k be a set for each $k \in K$. The cartesian product of the (indexed set of) sets X_k , $k \in K$ is the set

$$\prod_{k \in K} X_k = \left\{ x : K \rightarrow \bigcup_{k \in K} X_k \mid x(k) \in X_k \text{ for all } k \in K \right\}$$

and we write $x(k)$ as x_k . In the case that $K = \{1, \dots, n\}$, we write

$$\prod_{k \in K} X_k = \prod_{k=1}^n X_k = X_1 \times X_2 \times \dots \times X_n$$

In the case that $K = \mathbb{Z}^+ = \{1, 2, \dots\}$, we write

$$\prod_{k \in K} X_k = \prod_{k=1}^{\infty} X_k = X_1 \times X_2 \times X_3 \times \dots$$

In the case that $K = \{1, 2, \dots, n\}$ and $X_k = X$ for all $k \in K$, we also write

$$\prod_{k \in K} X_k = \prod_{k \in K} X = X \times X \times \dots \times X = X^n$$

In the case that $K = \mathbb{Z}^+ = \{1, 2, 3, \dots\}$ and $X_k = X$ for all $k \in K$, we also write

$$\prod_{k \in K} X_k = \prod_{k=1}^{\infty} X = X \times X \times X \times \dots = X^{\omega}$$

Remark

We remark that the n -tuple $x = (x_1, x_2, \dots, x_n) \in X_1 \times \dots \times X_n$ is (by definition) the function

$$x : \{1, 2, \dots, n\} \rightarrow \bigcup_{k=1}^n X_k$$

such that $x_k = x(k) \in X_k$ and that the sequence

$$x = (x_n)_{n \geq 1} = (x_1, x_2, x_3, \dots) \in X_1 \times X_2 \times X_3 \times \dots$$

is (by definition) equal to the function

$$x : \mathbb{Z}^+ \rightarrow \bigcup_{k=1}^{\infty} X_k$$

given by $x(k) = x_k$.

Notation

In the special case that X is a vector space, we also write

$$X^\infty = \{x \in X^\omega \mid x_k = 0 \text{ for all but finitely many } k \in \mathbb{Z}^+\}$$

In this case, both X^∞ and X^ω are both vector spaces. $\dim(\mathbb{R}^\infty) = \aleph_0$ and $\mathbb{R}^\omega = 2^{\aleph_0}$. $\mathbb{R}^\infty$ has standard basis $\{e_1, e_2, e_3, \dots\}$.

Example

When X_k is a set for each $k \in K$, for each $l \in K$, we have the projection map

$$p_l : \prod_{k \in K} X_k \rightarrow X_l$$

given by $p_l(x) = x_l = x(l)$. For any set Y , a function

$$f : Y \rightarrow \prod_{k \in K} X_k$$

determines, and is determined by, its component functions.

$f_l : y \rightarrow X_l$, where $f_l = p_l \circ f$, so $f_l(y) = p_l(f(y)) = f(y)_l = f(y)(l)$.

Definition 2.1.2

When X_k is a topological space for each $k \in K$, there are two commonly used topologies on $\prod_{k \in K} X_k$:

The *box topology* on $\prod_{k \in K} X_k$ is the topology generated by the basis of sets of the form

$$\prod_{k \in K} U_k \subseteq \prod_{k \in K} X_k$$

where each U_k is open in X_k .

The *product topology* on $\prod_{k \in K} X_k$ is the topology generated by the basis of sets consisting of the sets of the form $\prod_{k \in K} U_k$, where each U_k is open in X_k with $U_k = X_k$ for all but finitely many $k \in K$.

Note

The above two proposed bases are indeed bases of sets because

$$\left(\prod_{k \in K} U_k \right) \cap \left(\prod_{k \in K} V_k \right) = \prod_{k \in K} (U_k \cap V_k)$$

Also note that when K is finite, these two topologies are equal. When K is infinite, the box topology is finer than the product topology.

Theorem 2.1.3

Let $\mathcal{B}_k$ be a basis for X_k for each $k \in K$. The the set of sets of the form

$$\prod_{k \in K} B_k$$

where $B_k \in \mathcal{B}_k$ for all $k \in K$ is a basis for the box topology on $\prod_{k \in K} X_k$, where the set of sets of the form $\prod_{k \in K} B_k$, where $B_k \in \mathcal{B}_k \cup \{X_k\}$ for all $k \in K$ with $B_k = X_k$ for all but finitely many $k \in K$ is a basis for the product topology on $\prod_{k \in K} X_k$.

Proof: Exercise. □

Theorem 2.1.4

For each $k \in K$, let X_k be a subspace of Y_k (using the subspace topology). Then the box topology on $\prod_{k \in K} X_k$ is equal to the subspace topology on $\prod_{k \in K} X_k$ as a subspace of $\prod_{k \in K} Y_k$ using the box topology, and the product topology on $\prod_{k \in K} X_k$ is equal to the subspace topology on $\prod_{k \in K} X_k$ as a subspace of $\prod_{k \in K} Y_k$ using the product topology.

Theorem 2.1.5

Let Y be a topological space and let X_k be a topological space for each $k \in K$, and let $f : Y \rightarrow \prod_{k \in K} X_k$. Then when $\prod_{k \in K} X_k$ uses the product topology, f is continuous $\iff$ each component map $f_l : Y \rightarrow X_l$ (given by $f_l(y) = f(y)_l = f(\dots\dots\dots)$) is continuous.

Proof: ($\implies$) Suppose that f is continuous. Then (using either the box or product topologies on $\prod_{k \in K} X_k$) each projection map $p_l : \prod_{k \in K} X_k \rightarrow X_l$ is continuous because when $U \subseteq X_l$ is open, $p_l^{-1}(u) = \{x \in \prod_{k \in K} X_k \mid x_l = p_l(x) \in U\} = \prod_{k \in K} U_k$, where

$$U_k = \begin{cases} U & \text{if } k = l \\ X_k & \text{if } k \neq l \end{cases}$$

which is open in $\prod_{k \in K} X_k$ (using either the box or product topology). It follows that each component function f_l is continuous because $f_l = p_l \circ f$.

($\Leftarrow$) Suppose that each component map $f_l : p_l \circ f : Y \rightarrow \prod_{k \in K} X_k$ is continuous and that $\prod_{k \in K} X_k$ uses the product topology. To show that f is continuous, it suffices to show that $f^{-1}(B)$ is open in Y for every basic open set B in $\prod_{k \in K} X_k$.

Let B be a basic open set for the product topology on $\prod_{k \in K} X_k$. Say $B = \prod_{k \in K} U_k$, where each U_k is open in X_k with $U_k = X_k$ for all but finitely many indices $k \in K$. Let $L \subseteq K$ be the finite set of all indices $k \in K$ for which $U_k \neq X_k$. We have

$$\begin{aligned} f^{-1}(B) &= \left\{ y \in Y \mid f(y) \in \prod_{k \in K} U_k \right\} \\ &= \{y \in Y \mid f_k(y) = f(y)_k \in U_k \text{ for all } k \in K\} \\ &= \{y \in Y \mid f_k(y) \in U_k \text{ for all } k \in L\} \\ &= \bigcap_{k \in L} f_k^{-1}(U_k) \end{aligned}$$

which is open in Y since it is a finite intersection of open sets in Y (with $f_k^{-1}(U_k)$ is open in Y because U_k is open in X_k and $f_k : Y \rightarrow X_k$ is continuous). $\square$

Recall

$$\mathbb{R}^\infty \subseteq l_1 \subseteq l_p \subseteq l_q \subseteq l_\infty \subseteq \mathbb{R}^\omega$$

for $1 \leq p \leq q \leq \infty$. $\mathbb{R}^\infty$ denotes the set of eventually zero sequences.

Recall or verify that the p -norms on $\mathbb{R}^\infty$ (or on l_1) induce different topologies on $\mathbb{R}^\infty$ or (l_1) .

Question: do any of the p -norms induce the box or product topology on $\mathbb{R}^\infty \subseteq \mathbb{R}^\omega$?

Question: Is there a norm or metric on $\mathbb{R}^\omega$ which induces the box or product topology?

Also, we have the p -norms on $\mathbb{R}^n$. They all give the same topology on $\mathbb{R}^n$. More generally, when X is a finite dimensional vector space, all norms on X induce the same topology on X : when $L : X \rightarrow Y$ is a linear map between normed linear spaces, L is continuous $\Leftrightarrow \|L\|_{\text{op}} < \infty \Leftrightarrow L(\overline{B_X}(0, 1))$ is bounded in Y . When X is finite dimensional, $\overline{B_X}(0, 1)$ is compact and $L(\overline{B_X}(0, 1))$ is bounded, then L is continuous. In particular, when X is finite dimensional and $\|\cdot\|_1$ and $\|\cdot\|_2$ are two norms on X , then $I : (X, \|\cdot\|_1) \rightarrow (X, \|\cdot\|_2)$ is continuous and it is equal to its own inverse $I^{-1}(X, \|\cdot\|_2) \rightarrow (X, \|\cdot\|_1)$, which is continuous. So I is a homeomorphism. So for $U \subseteq X$, U is open in $(X, \|\cdot\|_1)$ if and only if U is open in $(X, \|\cdot\|_2)$.

Remark

Consequently, every finite dimensional vector space X has a standard topology. Pick any basis $\{u_1, \dots, u_n\}$, define

$$\left\langle \sum_{k=1}^n x_k u_k, \sum_{k=1}^n y_k u_k \right\rangle = \sum_{k=1}^n x_k y_k = x \cdot y$$

So the map $L : X \rightarrow \mathbb{R}^n$ given by

$$L\left(\sum_{k=1}^n x_k u_k\right) = \sum_{k=1}^n x_k e_k = x$$

is an inner product space isomorphism. Then use the inner product to define a norm, a metric and a topology. The resulting topology does not depend on the choice of basis.

2.2. Quotients Spaces**Definition 2.2.1**

Let X be a set and let $\sim$ be an equivalence relation on X . For $a \in X$, the *equivalence class* of a (under $\sim$) is the set

$$[a] = \{x \in X \mid a \sim x\}$$

Recall that distinct equivalence classes are disjoint and X is the (disjoint) union of the (distinct) equivalence classes. The set of all equivalence classes is denoted by

$$X / \sim$$

is called the *quotient (set) of X by $\sim$* .

$$X / \sim = \{[a] \mid a \in X\}$$

The map $q : X \rightarrow X / \sim$ given by $q(x) = [x]$ is called the *quotient map* (for $X / \sim$).

Definition 2.2.2

When X is a topological space, the *quotient topology* on $X / \sim$ is the topology obtained by stipulating that for $V \subseteq X / \sim$, V is open in $X / \sim \iff q^{-1}(V)$ is open in X . This gives a topology because

1. $q^{-1}(\emptyset) = \emptyset$
2. $q^{-1}(X / \sim) = X$
3. $q^{-1}\left(\bigcup_{k \in K} V_k\right) = \bigcup_{k \in K} q^{-1}(V_k)$
4. $q^{-1}\left(\bigcap_{k \in K} V_k\right) = \bigcap_{k \in K} q^{-1}(V_k)$

Note

When $V \subseteq X/\sim$, so V is a set of equivalence classes, then

$$\begin{aligned} q^{-1}(V) &= \{x \in X \mid q(x) \in V\} \\ &= \{x \in X \mid [x] \in V\} \\ &= \bigcup_{[x] \in V} [x] = \bigcup V \end{aligned}$$

Remark

For topological spaces X and Y ,

1. When $X \subseteq Y$ is a subset, the subset topology is the finest topology on X for which the inclusion map $\jmath : X \rightarrow Y$ (given by $\jmath(x) = x$) is continuous (for V open in Y , $\jmath^{-1}(V) = \{x \in X \mid \jmath(x) \in V\} = \{x \in X \mid x \in V\} = V \cap X$).
2. The product topology on $X \times Y$ is the coarsest topology on $X \times Y$ for which the two projection maps $f_x : X \times Y \rightarrow X$, $f_y : X \times Y \rightarrow Y$ are both continuous ($f_x^{-1}(U) = U \times Y$, $f_y^{-1}(V) = X \times V$).
3. When $\sim$ is an equivalence relation on X , the quotient topology on $X/\sim$ is the finest topology on $X/\sim$ for which the quotient map $q : X \rightarrow X/\sim$ is continuous.

Note

Let X be a set and let $\sim$ be an equivalence relation. Note that any function $g : X/\sim \rightarrow Y$ (where Y is any set) determines and is determined by a function $f : X \rightarrow Y$, which is constant on equivalence classes (meaning that for $x_1, x_2 \in X$, if $x_1 \sim x_2$, then $f(x_1) = f(x_2)$):
with g given by $g([x]) = f(x)$ and with f given by $f = g \circ q$, so
 $f(x) = g(q(x)) = g([x])$.

Theorem 2.2.3

Let X be a topological space. Let $\sim$ be an equivalence relation on X and let $f : X/\sim \rightarrow Y$. Let $g : X \rightarrow Y$ be the map given by $g(x) = f([x])$, that is, $g = f \circ q$. Then f is continuous $\iff g$ is continuous.

Proof: If f is continuous, then g is continuous because $g = f \circ q$, which is the composite of two continuous maps. Suppose that g is continuous. Let $V \subseteq Y$ be open. We need to show that $f^{-1}(V)$ is open in $X/\sim$. By definition of the quotient topology, $f^{-1}(V)$ is open in $X/\sim \iff q^{-1}(f^{-1}(V))$ is open in X . But $q^{-1}(f^{-1}(V)) = (f \circ q)^{-1}(V) = g^{-1}(V)$, which is open in X since g is continuous. $\square$

Definition 2.2.4

For a group G and set X , a *group action* of G on X is a function $\star : G \times X \rightarrow X$, which we write $\star(a, x)$ as $a \star x$, or simply as ax , such that

1. When $e \in G$ is the identity element, we have $e \star x = x$ for all $x \in X$.
2. For all $a, b \in G$ and all $x \in X$, we have $a \star (b \star x) = (ab) \star x$.

We say that G acts on X by the group action.

Remark

A group action of G on X determines and is determined by a group homomorphism $\rho : G \rightarrow \text{Perm}(X)$, where $\rho(a)(x) = a \star x$ (the homomorphism ρ is called a *representation* of G).

Given an action of G on X , we can define an equivalence relation on X by

$$x \sim y \Leftrightarrow y = a \star x \text{ for some } a \in G$$

In this case, the equivalence class of x is called the orbit of x (we might write $[x]$ as $\text{Orb}(x)$) and we write the quotient $X / \sim$ as $X \setminus G$. So

$$\begin{aligned} X \setminus G &= \{[x] \mid x \in X\} \\ &= \{\text{Orb}(x) \mid x \in X\} \end{aligned}$$

Example

For $\mathbb{S}^1 = \{u \in \mathbb{R}^2 \mid \|u\| = 1\}$, we have $\mathbb{S}^1 \times \mathbb{R}^1 \cong \mathbb{R}^2 \setminus \{0\}$.

Proof: Define $f : \mathbb{S}^1 \times \mathbb{R}^1 \rightarrow \mathbb{R}^2 \setminus \{0\}$ by

$$f(u, t) = e^t u$$

and define $g : \mathbb{R}^2 \setminus \{0\} \rightarrow \mathbb{S}^1 \times \mathbb{R}$ by

$$g(x) = \left(\frac{x}{\|x\|}, \ln(\|x\|) \right)$$

These maps are continuous (they are elementary functions) and they are inverses of each other. Indeed, $f(g(x)) = g(f(x)) = x$. □

Example

$\mathbb{S}^1$ acts on $\mathbb{R}^2 \setminus \{0\} = \mathbb{C}^*$ by (complex) multiplication. for $a \in \mathbb{R}^2 = \mathbb{C}$,

$$\text{Orb}(a) = [a] = \{ua \mid u \in \mathbb{S}^1\}$$

which is equal to the circle centered at 0 of radius $\|a\|$ (with $[0] = \{0\}$). Show that

$$\mathbb{R}^2 / \mathbb{S}^1 \cong [0, \infty) \subseteq \mathbb{R}$$

We define

$$f : \mathbb{R}^2 / \mathbb{S}^1 \rightarrow [0, \infty)$$

by $f([x]) = \|x\|$ and define

$$h : [0, \infty) \rightarrow \mathbb{R}^2 / \mathbb{S}^1$$

by $h(r) = [r] = \{re^{i\theta} \mid \theta \in \mathbb{R}\}$.

Note that $f : \mathbb{R}^2 / \mathbb{S}^1 \rightarrow [0, \infty) \subseteq \mathbb{R}$ is continuous because for the map $g : \mathbb{R}^2 \rightarrow [0, \infty) \subseteq \mathbb{R}$ given by $g(x) = \|x\|$, we have $g = f \circ q$. Since g is continuous (it's elementary), it follows that f is continuous. Also, it follows that h is continuous because $h = q \circ i$, where $i : [0, \infty) \rightarrow \mathbb{R}^2$ is the inclusion map $i(r) = (r, 0)$. Finally, we note that f and h are inverses by direct verification.